Homework 1

CS/ECE 374-B Intro to Algorithms & Models of Computation

University of Illinois Urbana-Champaign

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Maximum Points: 100
Due Date: Monday, February 03, 2025, 11:59 PM

Instructions

1. Answer **ALL** questions.
2. Show all the work.
3. Legibly scribe your solutions and upload them as individual problems on Gradescope.

Suggestions

While writing your answers and/or proofs, think about the following.

1. What is it that you are given? What is it that you are assuming? What is it that you are trying to prove?
2. How exactly does a step follow from the previous step? Assume nothing is apparent, even algebraic simplifications. Your reasoning should be clear and convincing.
3. If you are using a result from the class then state it clearly before using it. You should identify where exactly you need that result, cast your problem into the setup of the result, and then use it.
4. If you are using a result from outside the class then you should prove it before using it

Definitions

Definition 1. *If $(v_j, v_k) \in E$ then we say there is a* ***directed edge*** *from v_j to v_k: $v_j \to v_k$. If $u \to v$ is a directed edge then we call u a* ***parent*** *of v, and v a* ***child*** *of u.*

Definition 2. *In a directed graph G, we say the* ***degree*** *of a vertex is the number of edges connected to the node. In particular, the* ***in-degree*** *of a node is the number of edges coming into the vertex. Similarly, the* ***out-degree*** *of a node is the number of edges going out of the vertex.*

Definition 3. *In a directed graph, call a node* ***root*** *if it has in-degree 0. Similarly, call a node* ***leaf*** *if it has out-degree 0.*

Problems

1. Consider a directed graph with no cycles, a single root node and no parent having more than two children. Let the number of edges in a longest path from the root node to a leaf node be n. Prove that such a graph has at most $2^{n+1} - 1$ nodes. [20]

   Solution. We are given a directed graph with the following properties.

   - The graph has no cycles.
   - The graph has a single root node.
   - No parent in the graph has more than two children.
   - Any longest path from the root node to a leaf node has n edges. For simplicity, we define the number of edges in a longest path from the root node to a leaf node as the *height* of the graph.

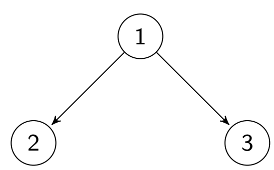
   We need to prove that the above graph has at most $2^{n+1} - 1$ nodes. Let $L(n)$ be the maximum possible number of nodes in the given graph of height n, we need to prove the following.

   $$L(n) = 2^{n+1} - 1.$$

   Note that as we need to prove a statement about *at most* number of nodes, we can consider the largest possible graph with the given properties.

   We can prove the given statement using induction as follows.

   i. *Base case.* When $n = 1$, we need to show that $L(1) = 2^{1+1} - 1 = 3$. When $n = 1$, the largest graph will appear as follows.

   The total number of nodes in the above graph is indeed 3. Hence, $L(1) = 3$. Hence, the statement holds for $n = 1$.

   ii. *Induction hypothesis.* The given statement is true for $n = k$, i.e.,

   $$L(k) = 2^{k+1} - 1. \tag{1}$$

   iii. *Induction Process.* Using the induction hypothesis given in Equation 1, we need to prove that the given statement holds true for $n = k + 1$, i.e.,

   $$L(k+1) = 2^{k+2} - 1. \tag{2}$$

   We start with the left-hand side of Equation 2 as follows.

   $$L(k+1) = L(k) + f(k),$$

   where $f(k)$ is the maximum additional number of nodes between a graph of height $(k+1)$ and k.
   As we have no parent having more than two children, a graph of height k will have a maximum of 2^k leaves. Hence, the maximum number of leaves in a graph with height $k+1$ will be 2×2^k due to the same reason that no parent has more than two children. This means that $f(k) = 2 \times 2^k = 2^{k+1}$. Hence, we have the following.

   $$\begin{aligned} L(k+1) &= L(k) + 2^{k+1}, && (3)\\ &= 2^{k+1} - 1 + 2^{k+1}, && (4)\\ &= 2(2^{k+1}) - 1,\\ &= 2^{k+2} - 1. \end{aligned}$$

   Equation 3 to 4 follows from the induction hypothesis given in Equation 1. Hence proved.

2. Show that every graph with no self-loops and with at least two nodes contains two nodes that have equal degrees. [20]

   Solution. We are given a graph that has at least two nodes and no self-loops. We need to prove that, the graph has two nodes that have the same degree.

   We will try to prove the given statement using proof by contradiction. The steps are as follows.

   (i). By way of contradiction, for the given graph, we assume that the given statement is false, i.e. we assume that no two nodes have the same degree. This is equivalent to saying that all nodes have distinct degrees.

   (ii). We next come up with a contradiction to the assumption made in (i). We first make the following observations for any graph with n nodes.

   - The least possible degree for a node is 0 when a node is not connected to any other node.
   - The highest possible degree for a node is $n-1$ when a node is connected to every other node.

   Using the above observations we can say that the degree of an arbitrary node v in the graph is always between 0 and $n-1$. Furthermore, following assumption (i), for the degrees of the n nodes to be distinct, they have to be $0, 1, 2, ..., n-1$. But it is impossible to have a node with degree 0 and another node with degree $n-1$ at the same time due to the following reason. A node having degree 0 implies that the node is not connected to any other node in the graph. Hence, we cannot have another node at the same time that has degree $n-1$, i.e. it is connected to every other node in the graph.

   This means that the distinct degrees n nodes can have are either $1, 2, ..., n-1$ or $0, 1, ..., n-2$, i.e. n nodes can have at the maximum $n-1$ distinct degrees, i.e all nodes can never have distinct degrees. Hence, we arrive at a contradiction.

   (iii). The assumption in (i) is incorrect. Hence, the given statement is true.

3. Prove or disprove the following statements. [20]

   (a) The sum of two rational numbers is rational.

   (b) The sum of a rational number and an irrational number is irrational.

   (c) The product of a non-zero rational number and an irrational number is rational.

   Solution.

   Lemma 1. *The Sum of two integers is also an integer.*

   Lemma 2. *The product of two integers is also an integer.*

   Lemmas 1-2 follow simply from the definition of an integer.

   Lemma 3. *The negative of a rational number is also a rational number.*

   Lemma 4. *The reciprocal of a rational number is also a rational number.*

   Lemmas 3-4 follow simply from the definition of a rational number.

   (a). We are given two rational numbers, and we need to check if their sum is rational or not. Let r_1 and r_2 be any two rational numbers, and s be the sum of these two rational numbers, i.e. $s = r_1 + r_2$. By definition of rational numbers, we know that there exist integers a, b, c and d such that $r_1 = \frac{a}{b}$ and $r_2 = \frac{c}{d}$. Hence, we have the following.

   $$\begin{aligned} s &= \frac{a}{b} + \frac{c}{d}, \\ &= \frac{ad + cb}{bd}. \end{aligned}$$

   Using Lemma 1 and Lemma 2, both $ad + cb$ and bd are integers. Hence, s is expressed as a ratio of two integers. Hence proved.

   (b). We are given a rational and an irrational number, and we need to check if their sum is irrational or not. Let r be a rational number and i be an irrational number, and s be the sum of these two numbers, i.e. $s = r + i$.

   We will try to prove the given statement using proof by contradiction. The steps are as follows.

(i). By the way of contradiction, assume that s is a rational number.

(ii). We try to come up with a contradiction of the assumption made in (i). We do the following.

$$\begin{aligned} s &= r + i, \\ \Rightarrow i &= s + (-r). \end{aligned}$$

Using Lemma 3, $-r$ is a rational number. Using (a), $s + (-r)$ is a rational number. Hence, i is a rational number. Hence, we arrive at a contradiction.

(iii). Thus the assumption in (i) is incorrect. Hence, the given statement is true.

(c). We are given a non-zero rational and an irrational number, and we need to check if their product is rational or not. Let r be a non-zero rational number and i be an irrational number, and p be the product of these two numbers, i.e. $p = r \times i$. By definition of rational numbers, we know that there exist integers a and b such that $r = \frac{a}{b}$. Hence, we have the following.

$$p = \frac{a}{b} \times i.$$

We will try to *disprove* the given statement using proof by contradiction. The steps are as follows.

(i). By the way of contradiction, assume that p is a rational number, i.e., there exist integers c and d such that $p = \frac{c}{d}$.

(ii). We try to come up with a contradiction of the assumption made in (i). We do the following.

$$\begin{aligned} p &= \frac{a}{b} \times i, \\ \Rightarrow \frac{c}{d} &= \frac{a}{b} \times i, \\ \Rightarrow i &= \frac{bc}{ad} \end{aligned}$$

Using Lemma 2, both bc and ad are integers. Hence, i is expressed as a ratio of two integers. Hence, i is a rational number. Hence, we arrive at a contradiction.

(iii). Thus the assumption in (i) is incorrect. Hence, the given statement is *false*.

Alternatively, we could disprove the given statement using a counterexample as follows. 5 is non-zero a rational number, and $\sqrt{2}$ is an irrational number but $5\sqrt{2}$ is not a rational number.

4. Prove that there are infinitely many prime numbers. [20]

Solution.

Definition 4. *A prime number is a natural number with exactly two distinct divisors: 1 and itself.*

We need to prove that there are infinitely prime numbers. We will try to prove this using proof by contradiction. The steps are as follows.

(i). By way of contradiction, assume that there is only a finite number of prime numbers. This is equivalent to saying that the set of all prime numbers is finite. Let $\{p_1, ..., p_n\}$ be that finite set of distinct prime numbers. Furthermore, without loss of generality, let $p_1 < \cdots < p_n$. Moreover, $p_1 = 2$, due to the fact that it is the smallest prime number.

(ii). We try to come up with a contradiction of the assumption made in (i). We do the following.

Consider the following number.

$$m = p_1 \times p_2 \times p_3 \times \cdots \times p_n + 1.$$

Clearly, $m > p_n$, and p_n is the largest prime number. Hence, m is not prime. But, due to the fact that every non-prime (composite) number has at least one prime factor (*prime factorization theorem*), m must be divisible by at least one of the primes in $\{p_1, \ldots, p_n\}$. In other words, there exists an integer i with $1 \leq i \leq n$ such that p_i divides m. Moreover, p_i also divides the product of all primes, i.e. $p_1 \times p_2 \times \cdots \times p_n$. Since p_i divides both m and the product of all the primes, it must also divide $m - p_1 \times p_2 \times p_3 \times \cdots \times p_n = 1$. Since $p_i \geq 2$, it is impossible that p_i divides one, Hence we have a contradiction.

(iii). Thus the assumption in (i) is incorrect. Hence the given statement, i.e. there are infinitely prime numbers, is true.

5. Give the recursive definition of the following languages. Explain why your solution is correct. [20]

   (a) A language that contains all strings.
   (b) A language that holds all the strings containing the substring 000.
   (c) A language that contains all palindrome strings using some arbitrary alphabet Σ.
   (d) A language that does not contain either three 0's or three 1's in a row. E.g., 001101 is in the language but 10001 is not.

Solution. Notice that by default we define the alphabet to be $\Sigma = \{0, 1\}$ unless stated otherwise. However, it is also acceptable if the answer is based on some arbitrary alphabet.

(a). We define a language L such that
   (i). $\epsilon \in L$
   (ii). If $w \in L$ and $a \in \Sigma$, $wa \in L$

(b). We define a language L such that
   (i). $\texttt{000} \in L$
   (ii). If $w \in L$ and $a \in \Sigma$, $wa \in L$ and $aw \in L$

(c). A string $w \in \Sigma^*$ is a palindrome if and only if:
   (i). $w = \epsilon$, or
   (ii). $w = \texttt{a}$ for some symbol $a \in \Sigma$, or
   (iii). $w = \texttt{a}x\texttt{a}$ for some symbol $\texttt{a} \in \Sigma$ and some palindrome $x \in \Sigma^*$ $(x \in L)$

(d). We are going to define two languages, L_1 and L_0 using a mutually recursive definition. L_1 contains all strings in L that start with 1, and L_0 contains all strings in L that start with 0. We are also going to have $\epsilon \in L_0$ and $\epsilon \in L_1$.

   Definition:
   (i). $\epsilon \in L_0$
   (ii). $\epsilon \in L_1$
   (iii). If $x \in L_0$, then $\texttt{1}x$ and $\texttt{11}x$ are in L_1
   (iv). If $x \in L_1$, then $\texttt{0}x$ and $\texttt{00}x$ are in L_0

   Then $L = L_1 \cup L_0$

   It is also acceptable to define $L_1 = \{\text{language that contains three 0's}\}$ and $L_2 = \{\text{language that contains three 1's}\}$, and $L = (L_1 \cap L_2)^c$